CHAYA छाया · system architecture

Two data streams in, one calibrated virtual patient, 2-hour spike prediction and what-if simulation out

1 · DATA STREAMS

Static / historical: EHR

FHIR R4 Bundle per patient

- Demographics, city, diet
- Diagnoses (SNOMED): T2D, HTN, NAFLD
- Labs (LOINC): HbA1c, FPG, lipids, eGFR
- Medications: metformin, glimepiride, DPP-4i, SGLT2i, insulin
- Genetics: TCF7L2 rs7903146

Dynamic / real-time: wearables

- CGM glucose, every 5 min
- Steps, heart rate
- Nightly HRV (RMSSD), sleep stages
- Meal log (Indian dishes, carbs, GI)
- Medication log

Synthetic cohort

150 Indian patients × 14 days · 604,800 sensor rows · physiology-driven generator. No real patient data.

2 · INGEST & HARMONISE

FHIR parser

LOINC / SNOMED codes → patient profile vector

Stream alignment

5-min grid · sensor gaps kept as gaps · meal and dose events placed on the timeline

Model inputs

Minute-level carbs, GI-dependent absorption, steps, sleep-debt and missed-dose modifiers

3 · DIGITAL TWIN CORE

Per-patient physiological model

`gut → plasma glucose ⇄ insulin action`
`extended Bergman minimal model`

1. **EHR prior.** BMI, waist, HbA1c, duration, TCF7L2, drugs → starting physiology
2. **Calibrate.** MAP fit to 7 days of CGM: insulin sensitivity, β-cell response, basal glucose, meal bioavailability, dawn effect
3. **Observe.** Runs beside live CGM; corrects its state and detects unlogged meals
4. **Habit model.** Learns when and how much this patient usually eats
5. **Forecast & what-if.** Simulates the next 2–4 h under routine or a doctor's scenario

↻ re-calibrates as new data arrives

4 · FUSION & PREDICTION

Fused feature vector

every 15 min

`18 EHR` `25 wearable` `11 twin`

Spike classifier

LightGBM · P(glucose > 180 mg/dL within 2 h)

Glucose regressor

LightGBM · glucose value at +2 h

Explanations

Tree SHAP grouped into clinical drivers: meal, routine, sleep, activity, medication, physiology

5 · DELIVERY

FastAPI on Railway

`PostgreSQL · /patients /simulate · /notes · /pipeline`

Chaya web app

- Home with live forecast slip
- Register, case sheet, stamps
- Reasons, twin physiology
- Rx pad what-if + case notes
- Product tour, case walkthrough

Patient nudge (next)

“Walk 15 min after lunch today”

Evaluation protocol

Patient-level split 105 train / 45 held-out. Temporal split: days 1–7 calibrate, days 8–14 predict. Threshold chosen on training folds only.

Privacy by design

Synthetic data only (DPDP Act 2023 / HIPAA sandbox rules). FHIR-native so it can plug into ABDM-consented records later.

Stack

Python · NumPy/Numba · SciPy · LightGBM · FastAPI · SQLAlchemy · PostgreSQL on Railway · vanilla JS + D3